Kyle Schwarber Hitting Project

Sam Kirkham, Jeremy Abbott

Our Research Question:

How well can pitch-level variables such as pitch location, pitch type, count, velocity, and pitcher handedness predict the probability that Kyle Schwarber gets a hit, and which of these are most influential?

Data

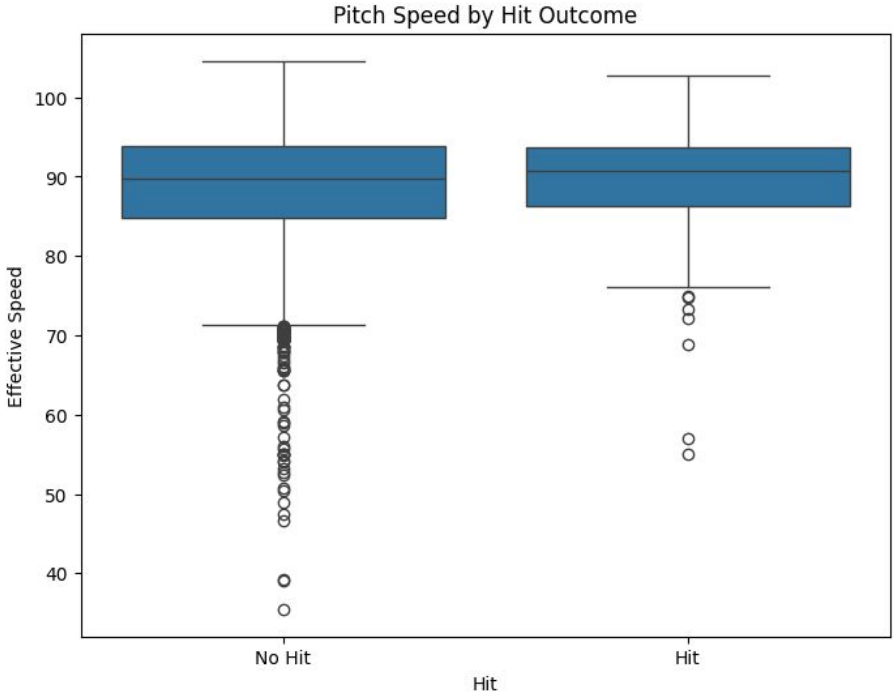
- Statcast powered by Google Cloud
- Official data hosted on BaseballSavant website
- Millions of data points from any level: season, game, at-bat, pitch
- Individual player data available as downloadable csv's

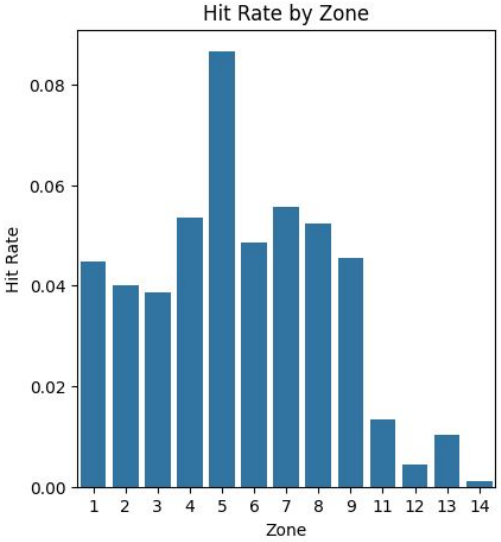
Our final dataset included pitch level statistics for Kyle Schwarber from the 2021-2025 seasons.

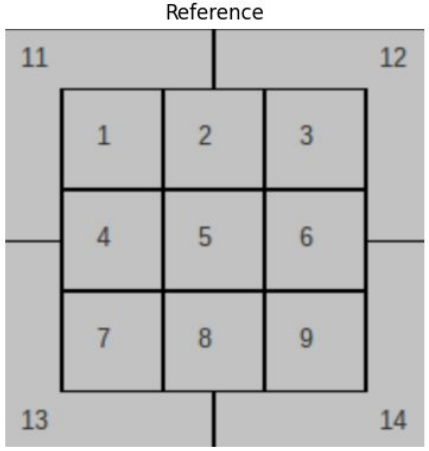
Data- Variable Definitions

- Pitch Zone: The zone the ball passes through as it goes over home plate
- Pitcher Handedness: Tells whether the pitcher is right or left handed
- Pitch Count: How many balls and strikes there are pre-pitch, allowing us to see the pitch situation
- Pitch Speed: The speed of the pitch as it leaves the pitcher's hand
- Pitch Spin: The spin rate on the ball as it leaves the pitcher's hand
- Pitch Number in AB: Number of pitches Kyle Schwarber has seen in this at bat
- Type of Pitch: What kind of pitch was thrown (fastball, curveball, slider, etc.)

EDA Insights







Data Imbalance

In our data we have about 14,000 pitches, of which we have about 640 hits

So our data is highly imbalance with less than 5% of the data being successes and 95% being fails

In order to deal with this imbalance we used SMOTE. We also avoided accuracy as our metric of choice since that would be heavily impacted by that class imbalance. Instead we focused on the ROC-AUC for our models

Methods & Hyperparameters

Random Forest

- # of trees, max depth, max features
- AUC: 0.668

SVM

- C (regularization strength), gamma
- AUC: 0.743

Logistic Regression

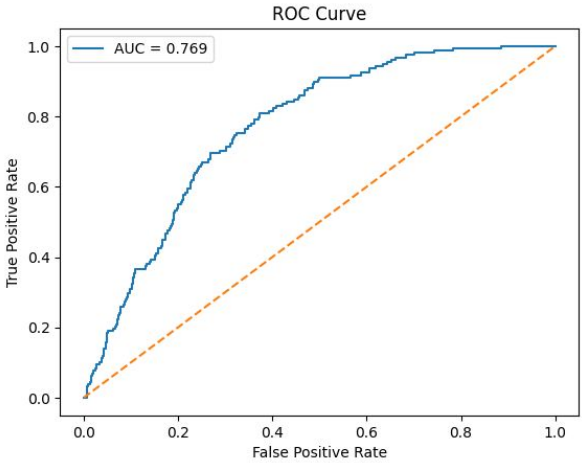
- C (regularization strength), gamma
- AUC: 0.769

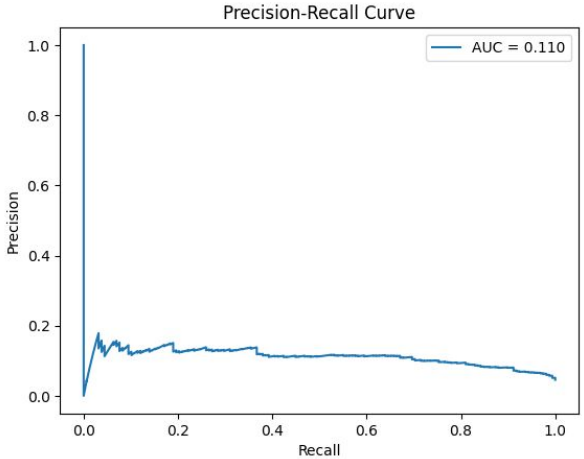
Validation- 75/25 train-test split with stratification

Results

This AUC value from our ROC curve of 0.769 shows that when the model predicts a “hit” about 77% of the time it actually was a “hit”

This AUC value from our Precision-Recall Curve is very low, however it is still better than random. In this curve an AUC equal to the class (In our data AUC of 0.05 would be a random model)

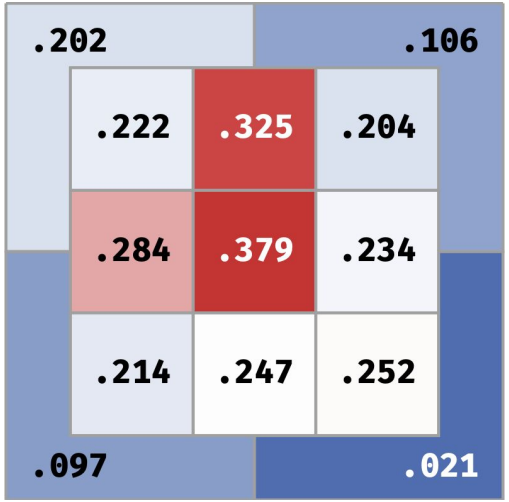




Permutation Importance

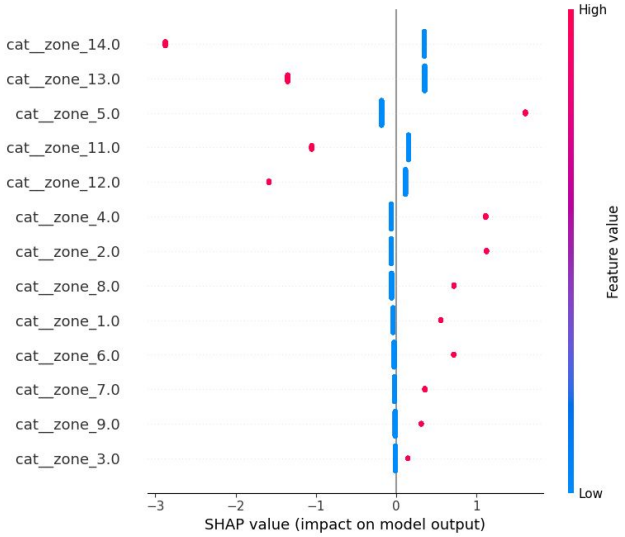
Permutation Importance mixes the value of that variable and then evaluates the model for the given metric and tells us how much it dropped, that is the importance value show.

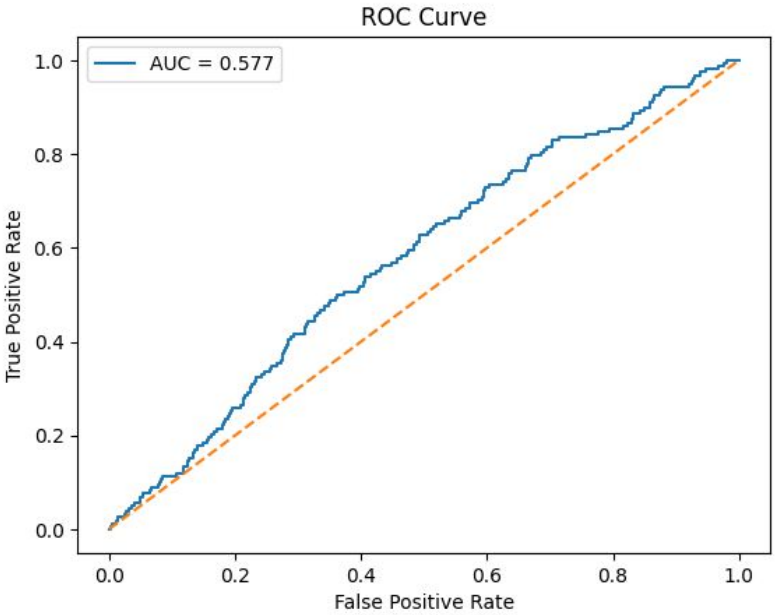
Feature	Importance
Zone	0.239
Pitch Count	0.030
Pitch Number	0.024
Pitch Name	0.016
Release Spin Rate	0.004
Effective Speed	0.003



Kyle Schwarber's actual batting average based on zone over the last 5 years

Effect of Zone





Ideas for Unsupervised Modeling

- K-means clustering to group similar pitch types and locations
- Identify clusters associated with a higher hit probability
- Use PCA to visualize pitch patterns
- Could help us see structure not captured by our supervised models

Conclusions

- Pitch level data can moderately predict hit outcomes
- *Where* the pitch is thrown matters more than *how* the pitch is thrown
- Logistic regression performed the best, suggesting relationship between predictors and outcome is mostly linear
- More complex models did not show any significant improvement
- Extreme class imbalance makes prediction more difficult